

VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VII	CS7T001	Data Science	3	0	0	3

Prerequisites for the course

1.	Basics Knowledge of Data Science and algorithm.
2.	Understand sorting, searching and hashing algorithms.

Prior Reading Material/useful links

1	https://www.google.co.in/books/edition/What_Is_Data_Science/-OQ2q5JqOdeC?hl=en&gbpv=1&dq=Data+Science&printsec=frontcover
2	https://www.google.co.in/books/edition/Data_Science_for_Business/4ZctAAAAQBAJ?hl=en&gbpv=1&dq=Data+Science&printsec=frontcover
3	https://www.google.co.in/books/edition/Getting_Started_with_Data_Science/b4YxCwAAQBAJ?hl=en&gbpv=1&dq=Data+Science&printsec=frontcover

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students shall be able to Build the fundamentals of data science.
2	CO2	Students shall be able to Apply Data Collection and Data Pre-processing Strategies.
3	CO3	Students shall be able to Compare and choose data visualization method for effective visualization of data
4	CO4	Students shall be able to Implement regression models, model evaluation and validation
5	CO5	Students shall be able to Test Multiple Parameters by using Grid Search

Syllabus:

Course Contents		Hours
Unit I	Introduction to Data Science: What is Data Science, importance of data science, Big data and data Science, The current Scenario, Industry Perspective Types of Data: Structured vs. Unstructured Data, Quantitative vs. Categorical Data, Big Data vs. Little Data, Data science process, Role Data Scientist.	7
Unit II	Data Collection and Data Pre-Processing: Data Collection Strategies, Data Pre-Processing Overview, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization.	7
Unit III	Exploratory Data Analytics : Descriptive Statistics, Mean, Standard Deviation, Skewness and Kurtosis, Box Plots, Pivot Table, Heat Map,	7

VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

	Correlation Statistics.	
Unit IV	Model Development : Simple and Multiple Regression, Model Evaluation using Visualization, Residual Plot, Distribution Plot, Polynomial Regression and Pipelines, Measures for In-sample Evaluation, Prediction and Decision Making, Feature Engineering	7
Unit V	Model Evaluation : Generalization Error, Out-of-Sample Evaluation Metrics, Cross Validation, Overfitting, Under Fitting and Model Selection, Prediction by using Ridge Regression, Testing Multiple Parameters by using Grid Search	7

Text Books

1.	Mitchell, Tom. M., "Data Science", McGraw-Hill Education, 1st Edition, May 2013.
2.	Segaran, Toby. "Programming Collective Intelligence- Building Smart Web 2.0 Applications", O'Reilly Media, August 2007.
3.	Miroslav, Kubat. "An Introduction to Data Science", Springer Publishing.

Reference Books

1.	Bishop, C. M., "Data Science and AI", Springer Publishing.
2.	Conway, Drew and White, John Myles, "Data Analytics", O'Reilly Media, February 2012.

Useful Links

1.	http://nptel.ac.in/courses/106105031/lecture by Dr. Debdeep Mukhopadhyay IIT Kharagpur
2.	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-033-computer-system-engineering-spring-2009/video-lectures/ lecture by Prof. Robert Morris and Prof. Samuel Madden MIT.
3.	https://www.google.co.in/books/edition/Applied_Cryptography_and_Network_Security/srsyEAAQBAJ?hl=en&gbpv=1&dq=cyber+security+%26+cryptography+notes&printsec=frontcover
4.	https://www.google.co.in/books/edition/Applied_Cryptography_and_Network_Security/OrBI6J5wHewC?hl=en&gbpv=1&dq=cyber+security+%26+cryptography+notes&printsec=frontcover